

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

FOR

B.TECH.

in

Artificial Intelligence and Machine Learning

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]



INSTITUTE OF ENGINEERING AND TECHNOLOGY
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Branch/Course: Artificial Intelligence and Machine Learning

Semester-I (First Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Basic Science Course	EPHY101	Engineering Physics	3+0
2.	Basic Science Course	EPHY151	Engineering Physics Lab	0+1
3.	Basic Science Course	EMAT103	Mathematics for Data Science	3+0
4.	Engineering Science Course	CSE101	Programming for Problem Solving	3+0
5.	Engineering Science Course	CSE151	Programming for Problem Solving Lab	0+1
6.	Engineering Science Course	ECE102	Basic Electrical Engineering	3+0
7.	Engineering Science Course	ECE152	Basic Electrical Engineering Lab	0+1
8.	Engineering Science Course	ME103	Fundamental of Mechanical Engineering and Mechatronics	3+0
9.	SEC Course	SEC-x	SEC Course	03
10.	AEC Course	AEC-x	AEC Course	02
11.	Induction Program	-----		-----
Total Credit				23

Semester-II (First Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Basic Science Course	ECHE101	Engineering Chemistry	3+0
2.	Basic Science Course	ECHE151	Engineering Chemistry Lab	0+1
3.	Basic Science Course	EMAT104	Mathematics for Machine Learning	3+0
4.	Humanities and Social Sciences including Management Course	HSM101	Professional Communication	3+0
5.	Humanities and Social Sciences including Management Course	HSM151	Professional Communication Lab	0+1
6.	Engineering Science Course	ECE101	Basic Electronics Engineering	3+0
7.	Engineering Science Course	ECE151	Basic Electronics Engineering Lab	0+1
8.	Professional Core Course	AIML102	Conversational AI	3+0
9.	SEC Course	SEC-x	SEC Course	03
10.	AEC Course	AEC-x	AEC Course	02
Total Credit				23



Basic Electrical Engineering

Course code		ECE102	
Category		Basic Engineering Course	
Course title		Basic Electrical Engineering (Theory)	
Scheme and Credits		Credits	3+0
Pre-requisites (if any)		-	
Course Objective:			
▪ To understand the electrical circuit fundamentals. ▪ Attain proficiency in analyzing single-phase AC circuits, exploring resonance phenomena, power factor concepts, and methods for power factor improvement. • To understand how the power supply and load are balanced, and how to measure power.			
Unit-1	Electrical Circuit Analysis: Introduction, Circuit Concepts, Concepts of network, Active and passive elements, Voltage and current sources, Concept of linearity and linear network, Unilateral and bilateral elements, Source transformation, Kirchhoff's laws, Loop and nodal methods of analysis, Star-delta transformation, AC fundamentals, Sinusoidal, square and triangular waveforms, Average and effective values, Form and peak factors, Concept of phasors, phasor representation of sinusoidally varying voltage and current.		09 (Lectures)
Unit-2	Network theorems (AC & DC with independent sources): Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum Power Transfer theorem (Simple numerical problems) Steady-State Analysis of Single-phase AC circuits: Analysis of series and parallel RLC Circuits, Concept of Resonance in series & parallel circuits, bandwidth and quality factor; Apparent, active & reactive powers, Power factor, Concept of power factor improvement and its improvement (Simple numerical problems).		09 (Lectures)
Unit-3	Three-phase AC circuit: Three phase system-its necessity and advantages, Star and delta connections, Balanced supply and balanced load, Line and phase voltage/current relations, three-phase power, and its measurement. Measuring instruments: Types of instruments, Construction and working principles of PMMC and moving iron type voltmeters & ammeters, Single phase dynamometer wattmeter, Use of shunts and multipliers (Simple numerical problems on shunts and multipliers), Single phase energy meter.		09 (Lectures)
Unit-4	Magnetic circuit: Concept of magnetic circuit, circuit analogy between electric and magnetic circuit, B-H curve, hysteresis and eddy current losses, Magnetic circuit calculation(Series and Parallel). Electrical Machines: Principle of operation, Construction, EMF equation, Phasor diagram, Equivalent circuit, Power losses, Efficiency (Simple numerical problems) of DC Machine, Single phase Transformer, Three phase Transformer, Single Phase Induction Machine , Three Phase Induction machine and Alternator. Power system: Basic concepts of electrical power generation, ac and dc transmission concepts, Models and performance of transmission lines and cables, Symmetrical and unsymmetrical fault analysis, Circuit breakers		09 (Lectures)

Text/Reference Books

1. Franklin F. Kuo, "Network Analysis and Synthesis," Wiley India Education, 2nd Ed., 2006.
2. Van, Valkenburg, "Network analysis," Pearson, 2019.
3. Sudhakar, A., Shyammoan, S. P., "Circuits and Network," Tata McGraw-Hill New Delhi, 1994.
4. A William Hayt, "Engineering Circuit Analysis," 8th Edition, McGraw-Hill Education.
5. A. Anand Kumar, "Network Analysis and Synthesis," PHI publication, 2019.

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

- CO1. Understand basics electrical circuits with nodal and mesh analysis.
- CO2. Appreciate electrical network theorems.
- CO3. Apply Laplace transform for steady-state and transient analysis.
- CO4. Determine different network functions.
- CO5. Appreciate the frequency domain techniques.

**FUNDAMENTALS OF MECHANICAL ENGINEERING AND MECHATRONICS**

Course code	ME103		
Category	Basic Engineering Course		
Course title	Fundamentals of Mechanical Engineering and Mechatronics		
Scheme and Credits	Credits	3+0	
Pre-requisites (if any)	-		
Course Objective:			
- To understand the basic concepts of the stresses and strains for different materials and strength of structural elements. - To study the behaviour of fluid at rest and in motion - To apply analytical techniques to the engineering problems and performance analysis of internal combustion engines, refrigerator and air-conditioner. - Understand key elements of Mechatronics system, representation into block diagram			
Unit-1	Introduction to Mechanics of Solid: Normal and shear Stress, strain, Hookes' law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety. Types of beams under various loads, Statically Determinate Beams, Shear force and bending moment in beams, Shear force and bending moment diagrams, Relationships between load, shear and bending moment		09 (Lectures)
Unit-2	Introduction to Fluid Mechanics and Applications: Fluids properties, pressure, density, dynamic and kinematic viscosity, specific gravity, Newtonian and Non-Newtonian fluid, Pascal's Law, Continuity Equation, Bernaulli's Equation and its applications, Basic Numerical problems.		09 (Lectures)
Unit-3	Introduction to IC Engines and RAC: IC Engine: Basic Components, Construction and Working of Two stroke and four stroke, SI & CI engine, merits and demerits, Introduction to electric, and hybrid electric vehicles. Refrigeration: Its meaning and application, unit of refrigeration; Coefficient of performance, methods of refrigeration, construction of domestic refrigerator, concept of heat pump. Formula based numerical problems on cooling load. Air-Conditioning: Its meaning and application, humidity, dry bulb, wet bulb, and dew point temperatures, comfort conditions, construction of window air conditioner.		09 (Lectures)
Unit-4	Introduction to Mechatronics: Evolution, Scope, Advantages and disadvantages of Mechatronics, Industrial applications of Mechatronics, Introduction to autotronics, bionics, and avionics and their applications. Sensors and Transducers: Types of sensors, types of transducers and their characteristics. Overview of Mechanical Actuation System		09 (Lectures)

Text/Reference Books

1. Basic Mechanical Engineering, G Shanmugam, S Ravindran, McGraw Hill
2. Basic Mechanical Engineering, M P Poonia and S C Sharma, Khanna Publishers
3. Mechatronics : Principles, Concepts and Applications, Nitaigour Mahalik, McGraw Hill
4. Mechanical Measurements & Control, Dr. D. S. Kumar. Metropolitan Book Company
5. Fluid Mechanics and Hydraulic Machines, Mahesh Kumar, Pearson India

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

- CO1.** Understand the concept of stress and strain, factor of safety, beams
- CO2.** Understand the basic component and working of internal combustion engines, electric and hybrid vehicles, refrigerator and heat pump, air-conditioning
- CO3.** Understand concept of mechatronics with their advantages, scope and Industrial application.
- CO4.** Apply concepts of strength of material for safe design, refrigeration for calculation of COP, concepts of fluid mechanics in real life.

Mathematics for Data Science (EMAT103)

Credit: 3+0

Syllabus

Course Objective: Students will be able to understand the basic mathematics for understanding the data science, and to use integration in the calculation of area, volume, mass, and center of gravity and to apply multivariable calculus to study the nature of multivariable functions. Students will finally understand the concept of Differential equation and its application.

1. Basic of Artificial Intelligence and Data Science, Set Theory, Number system, Sets and their operations, Relations and functions - Relations and their types, Functions and their types. Function of One variable, Function of two variables, Graphs and Tangents.
2. Calculus of functions of single variable: Review of limits, continuity, and differentiability. Mean value theorems: Rolle's Theorem, Lagrange's theorem, Cauchy's theorem, Taylor's theorem with remainders, indeterminate forms, curvature, curve tracing. Fundamental theorem of Integral calculus, mean value theorems of integral calculus, evaluation of definite integrals and various applications, Improper integrals: Beta and Gamma functions, differentiation under integral sign.
3. Calculus of Functions of Several Variables: Limit, continuity and differentiability of functions of several variables, partial derivatives and their geometrical interpretation, Tangent plane and normal line. Total differentiation, chain rules, Taylor's formula, maxima and minima, Lagrange's method of undetermined multipliers. Double and triple integrals, Jacobian, change of order of integration, change of variables, application to area, volumes, Mass, Centre of gravity.
4. Differential equation and its modelling with curve fitting: Modelling with Differential Equations, Direction Fields and Euler's Method, Linear and Bernoulli's differential equations, Nonlinear differential equations, Polar curves, angle between the radius vector and the tangent, angle between two curves. Pedal equations. Curvature and Radius of curvature - Cartesian, Parametric, Polar and Pedal forms.

Course Outcomes: At the end of course, the student will be able

C01: To analyze the nature (convergence or divergence) of a sequence or series.

C02: To apply mean value theorems in the study of motion of an object.

C03: To use integration in the calculation of area, volume, mass, and center of gravity.

C04: To apply multivariable calculus to study the nature of multivariable functions.

References

1. Kreyszig, E., Advanced Engineering Mathematics, John Wiley & Sons

Reference Books:

1. Piskunov, N., Differential and Integral calculus, Mir publishers Moscow (Vol. 1, Vol. 2)

**Engineering Mathematics-I**

Engineering Mathematics-I				
Course Code	EMAT-101			
Category	Basic Science Course			
Course Title	Engineering Mathematics-I (Theory) [Common to all B. Tech. Courses/Branches in I-Semester]			
Scheme and Credits	L	P	Credits	
	3	0	3	
Pre-requisites(if any)	Knowledge of Mathematics of 12 th class.			
Course objective	The objective of this course is to familiarize the graduate engineers with techniques in differential and integral calculus, vector calculus, trigonometry and matrix algebra. It aims to equip the students with standard concepts and tools from intermediate to advanced level that will enable them to tackle more advanced level of mathematics and applications that they would find useful in their disciplines.			
Unit-1	Differential Calculus: (Successive Differentiation) Leibnitz theorem and related problems, Partial derivatives, Homogenous functions, Euler's theorem for homogenous function, Total derivative, Change of variable, Curve tracing (Cartesian and Polar coordinates), Taylor and Maclaurin's theorems for a function of two variables, Jacobian, Maxima and Minima for functions of two variables, Lagrange multiplier method.			09(Lectures)
Unit-2	Integral Calculus: Beta and Gamma functions and their properties, Dirichlet's theorem and Liouville's theorem (without proof) and its applications. Multiple Integration: Double integrals and Triple integrals, change of order of integration in double integrals, Area as a double integral in Cartesian form. Trigonometry: Exponential functions, hyperbolic functions, logarithm of a complex number, general exponential function. Inverse Circular function of complex quantities, inverse hyperbolic functions.			09(Lectures)
Unit-3	Vector Calculus: Point function-scalar and vector point function. Vector differential operator, Gradient, Divergence and Curl, Normal on a surface, Directional Derivative, Second order differential operator, Laplacian operator. Vector Integration, Line integral, Circulation, Work done by a force, Surface integral, Volume integral, Gauss, Green, Stokes theorems (without proof) and related problems.			09(Lectures)
Unit-4	Matrix Algebra: Elementary operations on Matrices, Inverse of a Matrix by elementary operations, Rank of a Matrix, Echelon form of a Matrix, Normal form or Canonical form of a Matrix, Procedure to obtain Normal form $M = PAQ$ where P and Q are non-singular matrices. Consistency and Inconsistency of a system of linear homogeneous and non-homogeneous equations, Linear Dependence and Independence of vectors by rank method, Complex matrix, Conjugate of matrix, Transpose of conjugate of matrix, Hermitian matrix and Skew-Hermitian matrix, Orthogonal matrix, Periodic matrix, Idempotent matrix, Unitary matrix, Eigen values and Eigen vectors, Algebraic Multiplicity, Geometric Multiplicity, Regular eigen value, Caley-Hamilton theorem and its use in finding inverse of a matrix, Similarity transformation, Diagonalisation of square matrix, Power of matrix by Diagonalisation.			09(Lectures)

Suggested Text/Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, McGraw-Hill, Sixth Edition.
4. Veerarajan T., Engineering Mathematics, Tata McGraw-Hill, New Delhi, 2008.
5. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
6. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
7. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
8. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
9. Roger A. Horn, Charles R. Johnson, Matrix Analysis, Cambridge University Press.
10. H.K. Dass and Rama Verma: Engineering Mathematics: S. Chand Publications.

Programming for Problem Solving (CSE101)

Credit: 3+0

Course Objective: Students will be able to develop logics which will help them to create programs, applications in C also by learning the basic programming constructs they can easily switch over to any other language in future.

Syllabus

- 1. Introduction to Programming:** Introduction to components of a computer system, Algorithm: Representation of Algorithm, Flowchart, Pseudo code with examples, from algorithms to programs, source code. Programming Basics: Structure of C program: writing and executing the first C program, Syntax and logical errors in compilation, object and executable code. Components of C language: Standard I/O in C, Fundamental data types, Variables and memory locations, Storage classes. Arithmetic expressions & Conditional Branching: Arithmetic expressions and precedence: Operators and expression using numeric and relational operators, mixed operands, type conversion, logical operators, bit operations, assignment operator, operator precedence and associativity. Conditional Branching: Applying if and switch statements, nesting if and else, use of break and default with switch case.
- 2. Loops & Functions:** Iteration and loops: use of while, do while and for loops, multiple loop variables, use of break and continue statements. Functions: Introduction, types of functions, functions with array, passing parameters to functions, call by value, call by reference, recursive.
- 3. Arrays & Basic Algorithms:** Array notation and representation, manipulating array elements, using multi-dimensional arrays. Character arrays and strings, Structure, union, enumerated data types, Array of structures, passing arrays to functions. Basic Algorithms: Searching & Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, Notion of order of complexity.
- 4. Pointer & File Handling:** Pointers: Introduction, declaration, applications, Introduction to dynamic memory allocation (malloc, calloc, realloc, free), Use of pointers in self-referential structures, notion of linked list (no implementation) File handling: File I/O functions, Standard C pre-processors, defining and calling macros, command-line arguments.

Course Outcomes (CO): At the end of this course students will be able to:

CO1: To develop simple algorithms for arithmetic and logical problems.

CO2: To translate the algorithms to programs & execution (in C language).

CO3: To implement conditional branching, iteration and recursion.

CO4: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

CO5: To use arrays, pointers and structures to develop algorithms and programs.

Text Books:

1. The C programming by Kernighan Brain W. and Ritchie Dennis M., Pearson Education.